

## Bridge

Infrastructure: Bridge

Budget: \$10,00,000

Location: Dubai, UAE

Target Year: 2030

Risk Level: MEDIUM

Risk Score: 0.64/100

## Climate Resilience Report for Bridge Infrastructure in Dubai

Given the coordinates (25.1952, 55.2736), (25.1956, 55.2738), (25.1958, 55.2734), (25.1954, 55.2731) and the context of Dubai, this report outlines site-specific resilience strategies to mitigate flood hazards by 2030.

### STRATEGY 1

#### Elevation and Freeboard Enhancement

The bridge's structural components can be elevated to ensure a minimum freeboard of 0.5 meters above the projected flood levels, accounting for the 0.7m sea level rise by 2100. This strategy involves modifying the bridge's piers and abutments to increase the vertical clearance. By doing so, the bridge will be less susceptible to flooding, reducing the risk of damage and disruption.

### STRATEGY 2

#### Flood-Resistant Materials and Design

Implementing flood-resistant materials and design principles can enhance the bridge's resilience to flood hazards. This involves using materials that can withstand prolonged exposure to water, such as fiber-reinforced polymer (FRP) composites, and designing the bridge's components to minimize the risk of water ingress and damage. By adopting flood-resistant materials and design principles, the bridge can better withstand flood events and reduce the need for repairs.

### STRATEGY 3

#### Real-Time Monitoring and Early Warning Systems

A real-time monitoring and early warning system can be integrated into the bridge's infrastructure to detect rising water levels and alert authorities of potential flood events. This system can utilize sensors and IoT technologies to monitor water levels, rainfall intensity, and other environmental factors, providing critical information for proactive decision-making. By implementing such a system, authorities can take timely measures to protect the bridge and surrounding infrastructure, minimizing the risk of damage and disruption.

**RESILIENCE SCORE: 92%**



